

SYLLABUS :-

Pre-requisites: EC21001 or EC21003 **Introduction:** Scope and applications of analog electronic circuits. **Amplifier models:** Voltage amplifier, current amplifier, transconductance amplifier and transresistance amplifier. **Biasing schemes** for BJT and FET amplifiers, bias stability, various configurations (such as CE/CS, CB/CG, CC/CD) and their features, small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance etc., design procedure for particular specifications, low frequency analysis of multistage amplifiers. **High frequency transistor models,** frequency response of single stage and multistage amplifiers, cascode amplifier. **Various classes of operation** (Class A, B, AB, C etc.), their power efficiency and linearity issues. **Feedback topologies:** Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits, concept of stability, gain margin and phase margin. **Oscillators:** Review of the basic concept, Barkhausen criterion, RC oscillators (phase shift, Wien bridge etc.), LC oscillators (Hartley, Collpit, Clapp etc.), non-sinusoidal oscillators. **Current mirror:** Basic topology and its variants, V-I characteristics, output resistance and minimum sustainable voltage (VON), maximum usable load. **Differential amplifier:** Basic structure and principle of operation, calculation of differential gain, common mode gain, CMRR and ICMR. **OP-AMP design:** design of differential amplifier for a given specification, design of gain stages and output stages, compensation. **OP-AMP applications:** review of inverting and non-inverting amplifiers, integrator and differentiator, summing amplifier, precision rectifier, Schmitt trigger and its applications. **Active filters:** Low pass, high pass, bandpass and bandstop, design guidelines. **Digital-to-analog converters (DAC):** Weighted resistor, R-2R ladder, resistor string etc. **Analog-to-digital converters (ADC):** Single slope, dual slope, successive approximation, flash etc. **Switched capacitor circuits:** Basic concept, practical configurations, application in amplifier, integrator, ADC etc. **Other semiconductor devices:** UJT, SCR, diac, triac etc., device characteristics and application circuits. **Case study:** practical circuits of typical electronic systems